SYSTEM ANALYSIS & DESIGN

Sports Club Booking and Training Management System

Project Documentation

Document Information

Item	Description
Project	Sports Club Booking and Training Management System
System Type	Web-based system for one sports club
Primary Actors	Customer and Admin
Frontend	HTML, CSS, JavaScript
Backend	C# – ASP.NET Core Web API
Architecture Organization	MVC-style separation of concerns
Database	Microsoft SQL Server
ORM	Entity Framework Core
Payment	External gateway integration (Fawry or Paymob to be selected)
Date / Version	November 2026 / Version 1.0

Documentation note: This document reflects the requirements confirmed so far. Items that still require implementation decisions are explicitly marked as assumptions or open decisions.

1. System Analysis

1.1 System Overview

The Sports Club Booking and Training Management System is a web-based platform for a single sports club. Customers can reserve available football fields for one or more consecutive hours and book group training sessions. The club administrator manages fields, schedules, coaches, training sessions, bookings, payments, and reports through an administration area.

1.2 Problem Statement

Managing field reservations and group training sessions manually or through disconnected methods can cause scheduling conflicts, unclear availability, difficulty tracking payments, and limited visibility into booking performance. The proposed system centralizes these activities, validates availability, tracks payment status, and provides administrative reports.

1.3 Objectives

- Allow customers to view field types and available time slots.
- Allow customers to reserve multiple consecutive hours in one booking.
- Allow customers to browse and reserve group training sessions.
- Allow the Admin to manage fields, prices, schedules, coaches, and sessions.
- Prevent overlapping field reservations and training capacity overbooking.
- Create a separate payment transaction for each booking and verify it automatically through the payment gateway.
- Cancel unpaid pending bookings automatically after a configurable payment period.
- Provide the Admin with booking and revenue reports.

1.4 Scope

In Scope

- Customer registration, login, and booking history.
- Admin authentication and full administrative management.
- Field types, field records, prices, and availability schedules.
- Field reservations for multiple consecutive hours.
- Group training sessions with a coach, schedule, duration, price, and maximum capacity.
- Training session reservations and participant-capacity validation.
- Payment transaction creation, gateway verification, and booking status updates.
- Automatic expiry and cancellation of unpaid bookings.
- Administrative booking and revenue reports.

Out of Scope

- Managing more than one sports club.
- Coach self-registration or coach accounts.
- Customer-initiated booking cancellation.
- In-app card or wallet processing; payment is handled by an external gateway.
- Notifications, email alerts, and SMS alerts in the current version.
- Searching across different clubs or external venues.

1.5 Actors

Actor	Description	Main Permissions
Customer	A person using the website to book club services.	Register/login, view field availability, book fields, view group sessions, book training, view own bookings.
Admin	The club employee responsible for the system.	Manage fields, field schedules, prices, coaches, training sessions, bookings, payment records, and reports.

The payment gateway is an external system integration, not a human user role.

1.6 Functional Requirements

ID	Functional Requirement
FR-01	The system shall allow customers to create an account and log in.
FR-02	The system shall authenticate Admin users and restrict administrative functions to authorized users.
FR-03	The system shall display the field types and fields configured by the Admin.
FR-04	The system shall display available dates and time slots for a selected field.
FR-05	The system shall allow a customer to select one or more consecutive hours.
FR-06	The system shall calculate the booking price using the price assigned to the field type and selected duration.
FR-07	The system shall validate availability before creating a field booking.
FR-08	The system shall create a booking with Pending Payment status before payment is completed.
FR-09	The system shall create a unique payment transaction associated with the relevant booking.
FR-10	The system shall redirect the customer to the external payment provider.
FR-11	The system shall verify payment results through a secure server-side gateway integration/webhook.
FR-12	The system shall change a booking to Confirmed only after successful payment verification.
FR-13	The system shall automatically cancel an unpaid booking after its payment deadline.
FR-14	The system shall prevent overlapping active bookings for the same field and time.
FR-15	The system shall allow customers to view their own booking records and statuses.
FR-16	The system shall display available group training sessions.
FR-17	The system shall allow customers to reserve a group training session.
FR-18	The system shall prevent the number of confirmed training participants from exceeding the session capacity.
FR-19	The system shall manage a separate payment transaction for each training booking.
FR-20	The Admin shall create, update, and delete field records and field types as permitted by booking history constraints.
FR-21	The Admin shall manage field availability schedules and time slots.
FR-22	The Admin shall add, update, and remove coach information.
FR-23	The Admin shall create and manage group training sessions.
FR-24	The Admin shall view and manage booking and payment records.
FR-25	The Admin shall view booking and revenue reports.

1.7 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Usability	The interface should be clear and easy to use for customers and Admin users.
NFR-02	Security	Passwords must be stored as secure hashes; protected endpoints must require authorization.
NFR-03	Payment Security	Payment status must be verified on the server using the gateway's secure verification mechanism; a browser redirect alone is not proof of payment.
NFR-04	Data Integrity	The system must preserve consistent relationships between users, bookings, time slots, sessions, and payments.
NFR-05	Concurrency	The system must prevent two customers from successfully reserving the same field time slot.
NFR-06	Reliability	Booking and payment updates should use safe database transactions and handle repeated gateway notifications idempotently.
NFR-07	Compatibility	The website should work on current major desktop and mobile browsers.
NFR-08	Maintainability	The code should separate controllers, services, data access, and domain models.

NFR-09	Validation	The system must validate user input and reject invalid dates, times, prices, and capacity values.
NFR-10	Authorization	Only authorized Admin users may perform administrative operations.
NFR-11	Performance	Common pages and booking operations should respond within an acceptable time under expected club usage.
NFR-12	Auditability	Important booking and payment status changes should be traceable.

1.8 Business Rules

1. A customer cannot cancel a booking through the current system.
2. A field booking remains Pending Payment until the gateway confirms successful payment.
3. An unpaid booking is automatically cancelled when its payment deadline expires.
4. Each booking must have its own payment transaction/reference, even when the same external provider is used.
5. A field booking may contain multiple consecutive hourly slots.
6. The same field time cannot be allocated to more than one active booking.
7. Field pricing is determined by field type; the Admin manages the price for each field type.
8. Training sessions are group sessions and have a maximum participant capacity.
9. Coaches do not create accounts; the Admin manages coach records.
10. Only the Admin manages the club's fields, coaches, schedules, and training sessions.

1.9 Assumptions and Open Decisions

- The payment provider (Fawry or Paymob) has not yet been selected. The integration details and supported webhook mechanism must be confirmed from the selected provider's official technical documentation.
- The payment deadline duration is not yet specified. It should be configurable by the Admin or system configuration.
- The current requirement says the customer chooses field type and time. The exact rule for assigning a specific physical field when multiple fields share one type must be finalized.
- The payment approach was updated from a shared fixed link to a unique transaction/payment reference per booking to support automatic verification. A shared generic link alone is not sufficient for reliable automatic matching.
- Revenue reports should define whether revenue means successfully paid transactions only and how refunds/failed payments are treated. Refund handling is currently outside scope.

2. Use Case Model

The system has two human actors: Customer and Admin. The external payment gateway participates in payment processing by returning a verified transaction result.

Use Case Diagram

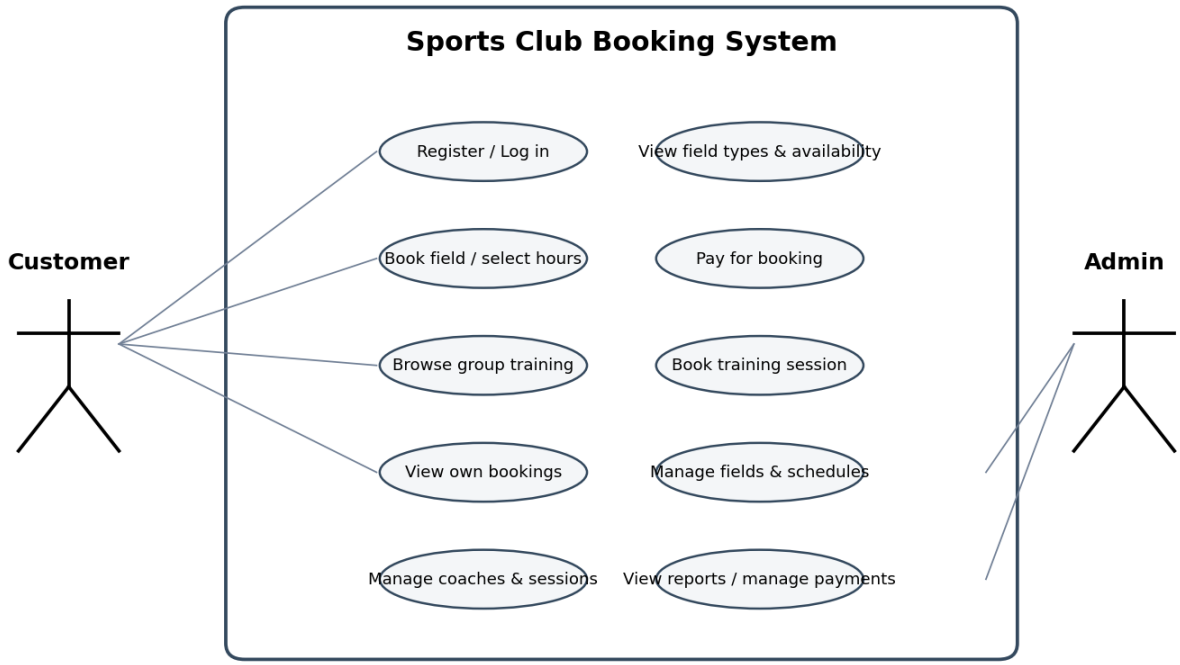


Figure 1. Use Case Diagram (high-level)

2.1 Use Case Descriptions

Use Case	Actor(s)	Main Flow	Outcome
UC-01 – Register / Log in	Customer	The customer creates an account or enters valid credentials to access the system.	A valid customer account exists or a new account is created.
UC-02 – Book a Field	Customer	The customer selects a field type, date, available time, and one or more consecutive hours. The system validates availability and creates a Pending Payment booking.	A pending booking and a unique payment transaction are stored.
UC-03 – Pay for Field Booking	Customer, Payment Gateway	The customer is redirected to the selected payment provider. The provider processes payment and sends a server-side result.	A verified successful payment changes the booking to Confirmed.
UC-04 – Book Group Training	Customer	The customer views available group sessions, checks session details and capacity, then submits a booking.	A pending training booking is created with its own payment transaction.
UC-05 – Pay for Training	Customer, Payment Gateway	The customer completes payment for the selected training session.	The training booking is confirmed only after successful verification.
UC-06 – Manage Fields and Schedules	Admin	The Admin creates or updates fields, field types, prices, and available schedules.	Updated field and availability data is stored.
UC-07 – Manage Coaches and Sessions	Admin	The Admin maintains coach details and creates group training sessions with schedule, duration, price, and capacity.	The session becomes available for customer booking when published/active.
UC-08 – View Reports	Admin	The Admin selects a reporting period and views booking and revenue summaries.	The system displays report data based on stored records.

2.2 Use Case Notes

The diagrams are high-level representations. Detailed alternate flows (invalid credentials, unavailable time, full training session, failed payment, duplicated webhook, and expired payment) should be included in the implementation test cases.

3. Software Architecture

3.1 Architecture Style

The proposed solution uses a layered ASP.NET Core Web API organized with MVC-style separation of concerns. The browser-based frontend communicates with API controllers over HTTPS. Controllers receive requests, application services apply business rules, Entity Framework Core accesses SQL Server, and a payment integration component communicates with the external payment gateway.

- Presentation layer: HTML, CSS, and JavaScript in the browser.
- API / Controller layer: ASP.NET Core Web API controllers and request/response models.
- Business layer: booking, availability, training, payment, and reporting services.
- Data access layer: Entity Framework Core DbContext and repositories/data access services where appropriate.
- Database layer: Microsoft SQL Server.
- External integration: payment provider API and webhook endpoint.

3.2 Technology Stack

Layer	Technology	Purpose
Frontend	HTML5, CSS3, JavaScript	User interface and client-side interactions.
Backend	C# / ASP.NET Core Web API	Business logic and HTTP endpoints.
Architecture	MVC-style separation	Organizes controllers, models, and application responsibilities.
Database	Microsoft SQL Server	Stores users, fields, schedules, bookings, sessions, and payments.
ORM	Entity Framework Core	Maps C# entities to database tables and performs data access.
Payment Integration	Fawry or Paymob (TBD)	Processes transactions and returns payment status.
Transport	HTTPS / JSON	Secure communication between browser, API, and payment provider.

3.3 Architecture Diagram

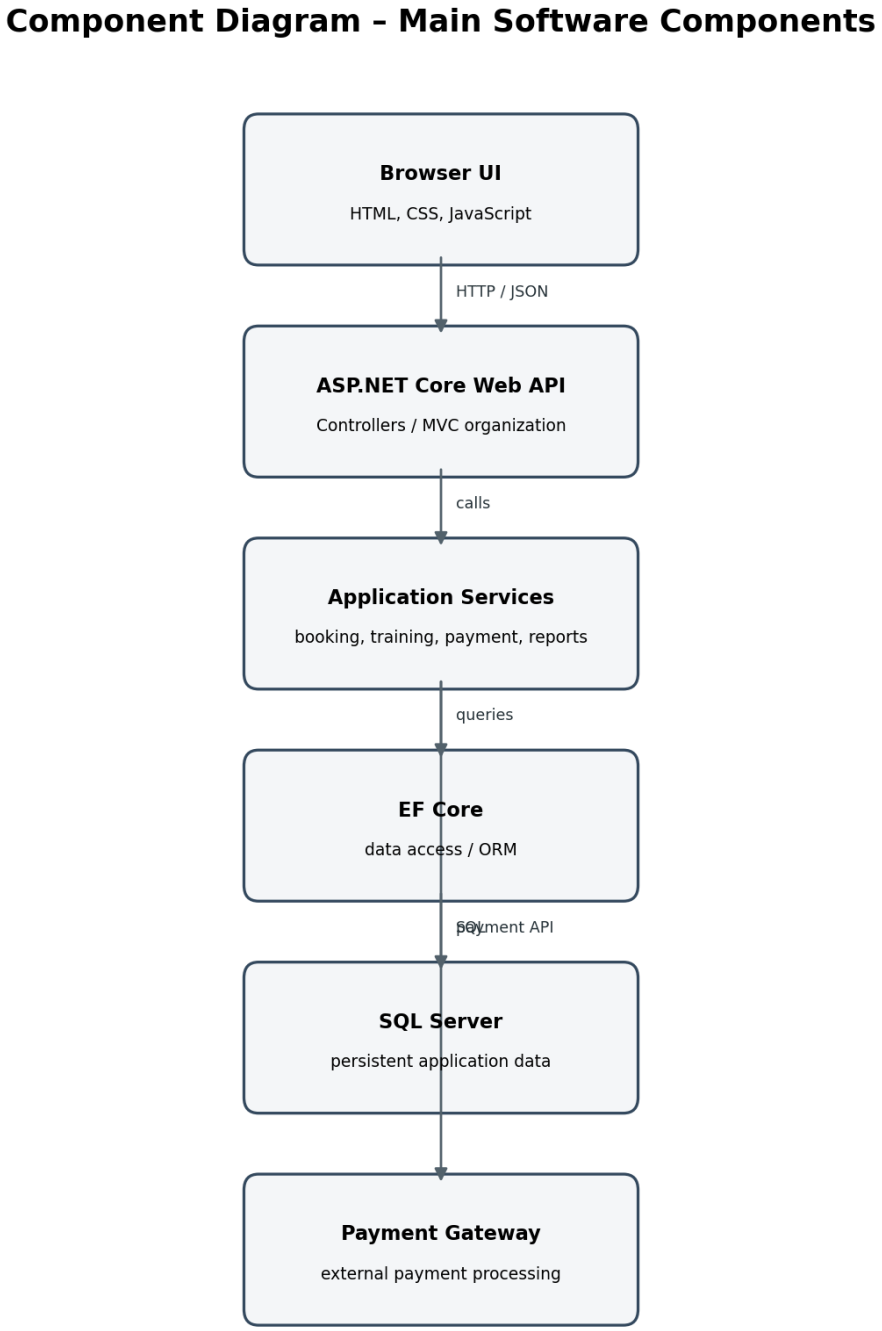


Figure 2. High-level component interaction

4. Database Design and Data Modeling

The submitted ERD is included below as the team's current conceptual model. The proposed logical schema that follows clarifies keys and recommends a few changes needed for reliable date-based booking and automatic payment verification.

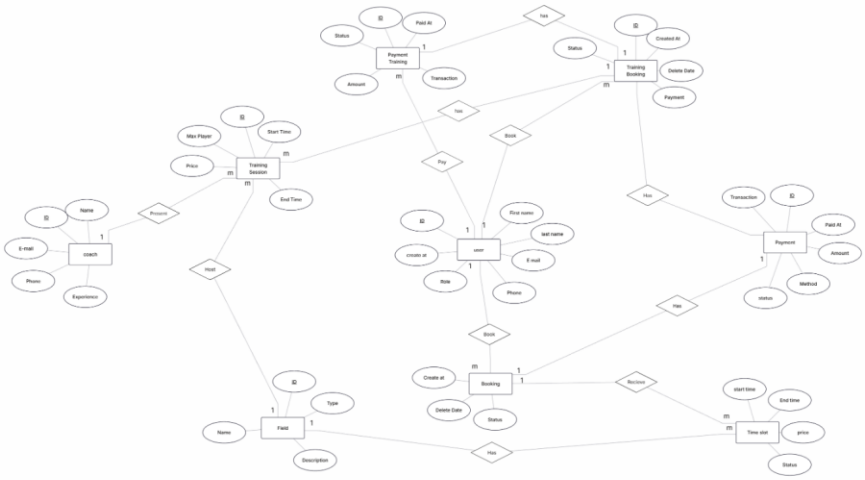


Figure 3. Original ERD

4.1 ERD Review and Recommendations

- Add a booking date to the time-slot/availability design. StartTime and EndTime alone do not identify a real calendar reservation.
- Represent a field booking's selected hours explicitly. A BookingTimeSlot junction table is recommended if one booking can include multiple hourly slots.
- Use one Payment table for both field and training bookings if the implementation team can enforce that each payment belongs to exactly one booking type. Alternatively, use separate FieldPayment and TrainingPayment tables consistently; do not duplicate payment data in both a booking attribute and a payment entity.
- Use a unique gateway transaction/reference per booking and store gateway status, amount, method/provider, creation time, and paid time.
- Add ExpiresAt to pending bookings so automatic cancellation can be implemented reliably.
- TrainingSession should contain CoachId, StartAt, EndAt or Duration, Price, MaxPlayers, and an active/status field.
- TrainingBooking should link CustomerId to TrainingSessionId and store status and creation time.
- Store password hashes, never plain-text passwords.

4.2 Logical Schema

Table	Suggested Columns	Purpose
Users	UserId (PK), FirstName, LastName, Email (Unique), Phone, PasswordHash, Role, CreatedAt	Customer and Admin accounts.
FieldTypes	FieldTypeId (PK), TypeName, HourlyPrice, Description	Defines field category and its hourly price.
Fields	FieldId (PK), FieldTypeId (FK), Name, Description, IsActive	Physical fields owned by the one club.
TimeSlots	TimeSlotId (PK), FieldId (FK), SlotDate, StartTime, EndTime, Status	A dated availability slot for a specific physical field.
Bookings	BookingId (PK), UserId (FK), CreatedAt, Status, ExpiresAt, TotalAmount	A customer's field booking.
BookingTimeSlots	BookingId (PK/FK), TimeSlotId (PK/FK)	Associates one booking with one or more consecutive slots.
Coaches	CoachId (PK), Name, Email, Phone, Experience, IsActive	Coach details managed by Admin.
TrainingSessions	SessionId (PK), CoachId (FK), StartAt, EndAt, Duration, Price, MaxPlayers, Status	A scheduled group training session.
TrainingBookings	TrainingBookingId (PK), UserId (FK), SessionId (FK), CreatedAt, Status, ExpiresAt, TotalAmount	Customer registration for a group session.
Payments	PaymentId (PK), BookingId (FK, nullable), TrainingBookingId (FK, nullable), Provider, GatewayReference (Unique), Amount, Method, Status, CreatedAt, PaidAt	One transaction per field or training booking; enforce exactly one booking reference.

4.3 Relationships and Cardinalities

Relationship	Cardinality	Explanation
User – Booking	1 : Many	One customer can create many field bookings; each booking belongs to one customer.
FieldType – Field	1 : Many	One field type can describe many physical fields.
Field – TimeSlot	1 : Many	One field has many dated availability slots.
Booking – BookingTimeSlot	1 : Many	A booking can contain multiple hourly slots.
TimeSlot – BookingTimeSlot	1 :1	A time slot can be linked to at most one active booking.
Coach – TrainingSession	1 : Many	One coach can conduct many training sessions.
TrainingSession – TrainingBooking	1 : Many	One session can have multiple customer bookings, up to MaxPlayers.
User – TrainingBooking	1 : Many	One customer can book multiple training sessions.
Booking – Payment	1: 1	A booking has at most one successful payment transaction in the simple model; retries may require multiple payment attempts.
TrainingBooking – Payment	1: 1	A training booking has at most one successful payment transaction in the simple model.

4.4 Normalization

The proposed schema aims to satisfy Third Normal Form (3NF): each table represents one main subject, non-key attributes depend on the key, and repeated coach, field type, and customer information is stored once and referenced by foreign keys. BookingTimeSlots resolves the many-to-many association between bookings and selected slots. Payment attempts may be modeled as multiple rows if the gateway allows retries.

4.5 Important Database Constraints

- Users.Email must be unique.
- Payments.GatewayReference must be unique.
- StartTime must be earlier than EndTime; slot dates and session dates must be valid.
- MaxPlayers must be greater than zero; confirmed participants must not exceed MaxPlayers.
- Field slot allocation must be protected against concurrent double booking using database constraints/transactions and server-side validation.
- A payment must reference exactly one of BookingId or TrainingBookingId.
- Amounts and prices must be non-negative and stored using an appropriate decimal type.

5. Data Flow and System Behavior

5.1 Context-Level DFD

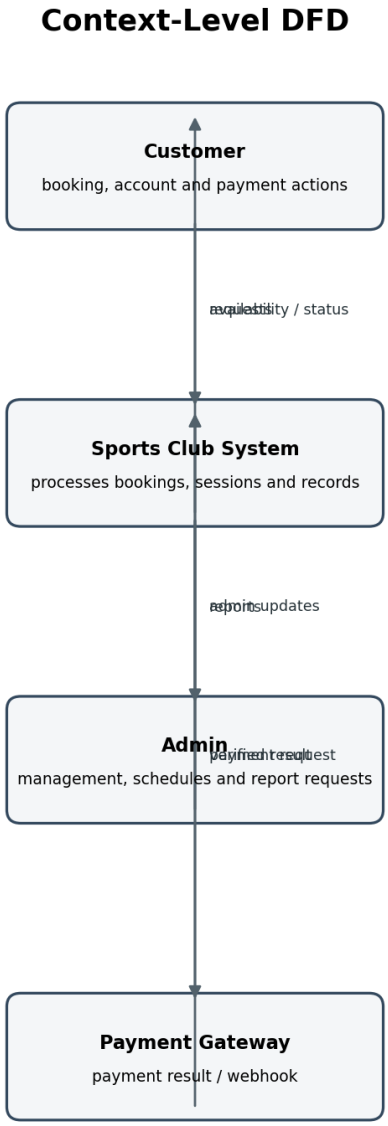


Figure 4. Context-Level Data Flow Diagram

5.2 Level 0 DFD

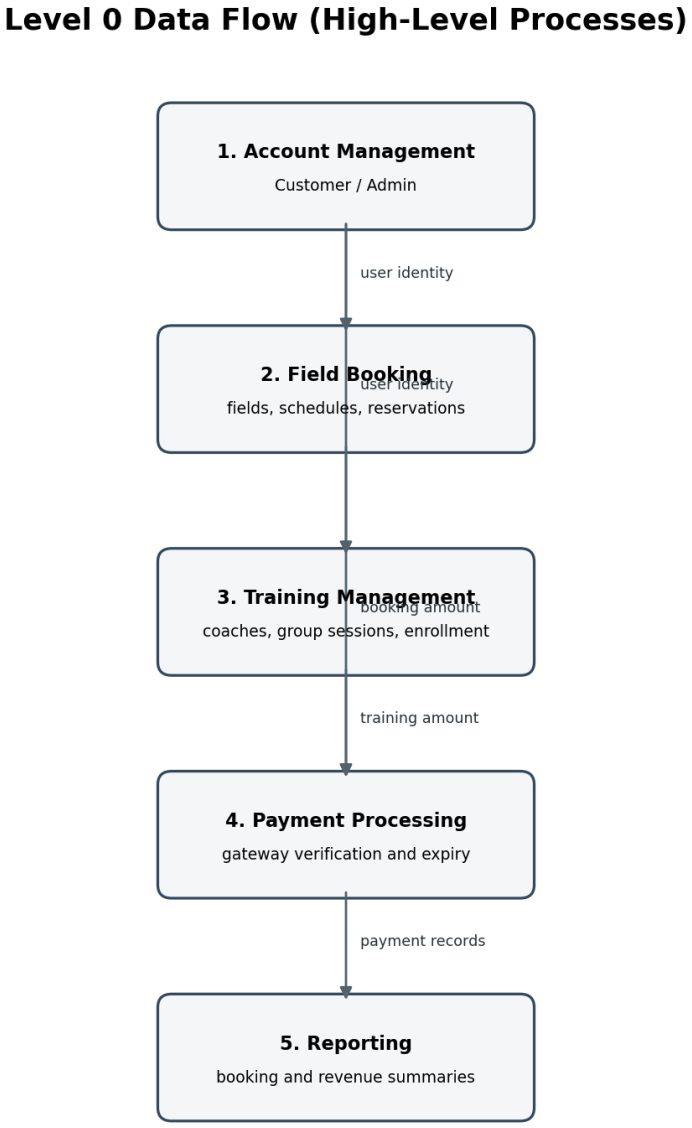


Figure 5. Level 0 DFD – Main System Processes

The main processes are account management, field booking, training management, payment processing, and reporting. These processes read and update the system's persistent records.

5.3 Sequence Diagram – Field Booking and Payment

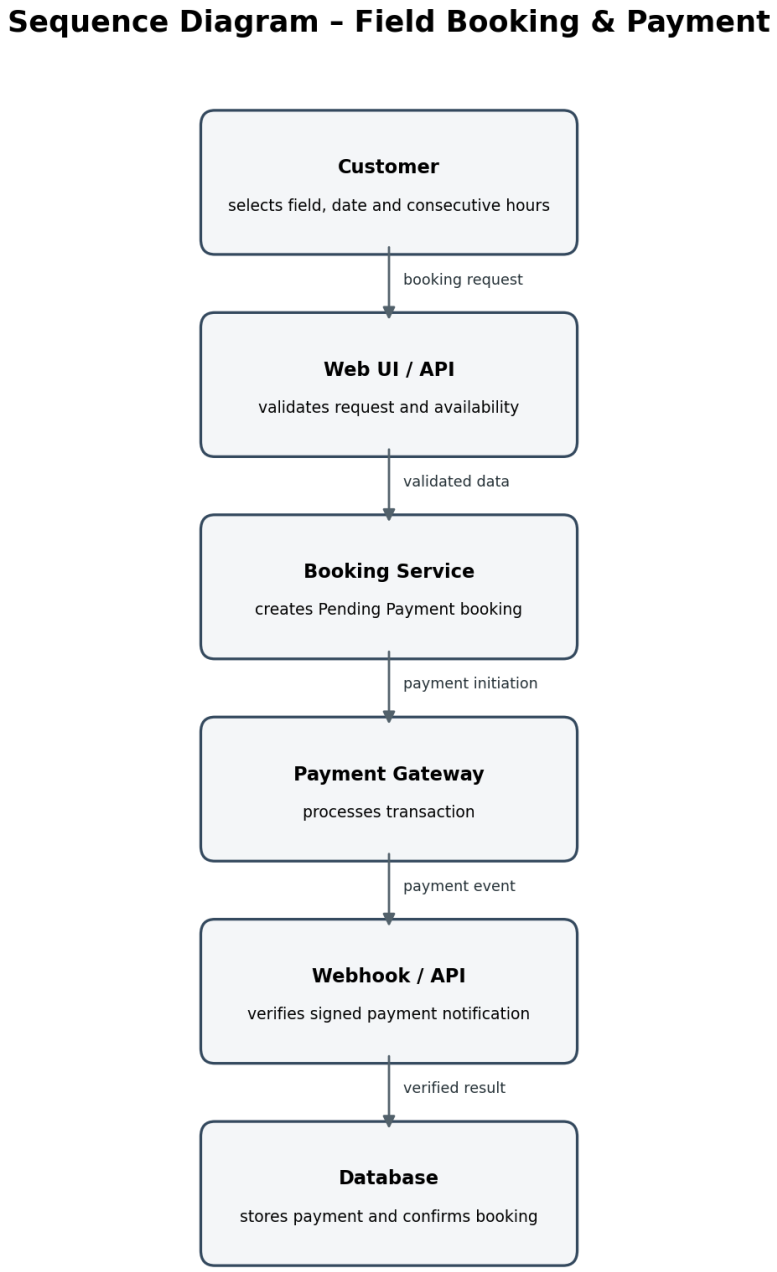


Figure 6. Sequence Diagram – Field Booking and Payment Verification

The payment webhook must be authenticated/validated according to the selected provider's specification. The system should verify the transaction server-side, check the amount and booking reference, and process repeated webhook events safely.

5.4 Activity Diagram

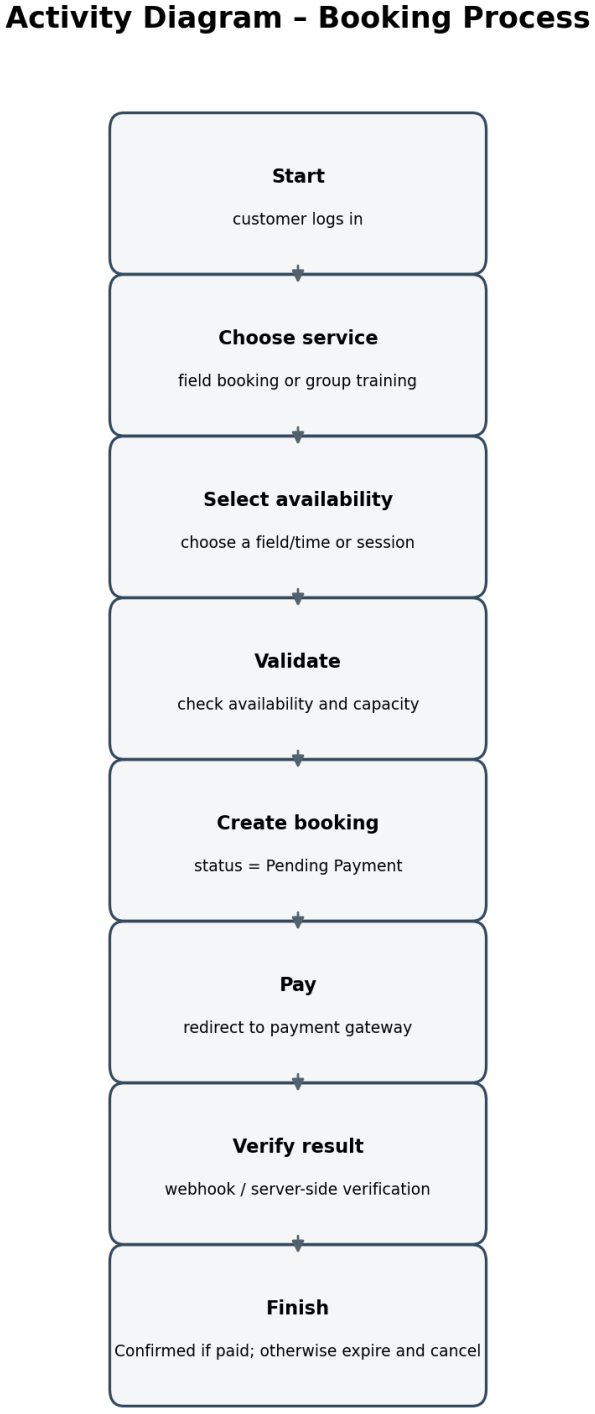


Figure 7. Activity Diagram – Booking Workflow

5.5 State Diagram

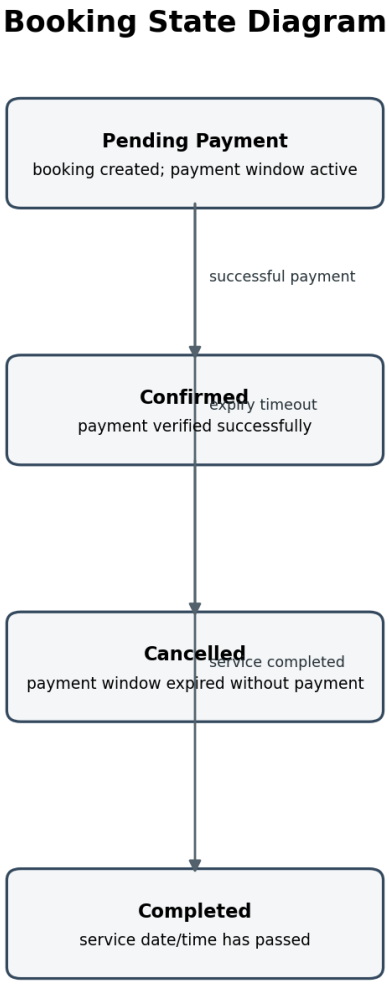


Figure 8. Booking State Diagram

The diagram shows the main booking lifecycle. A separate Failed or Payment Failed status may be added if the team needs to distinguish a declined payment from a booking that simply expired. A completed state is reached after the booked service has taken place.

5.6 Class Diagram

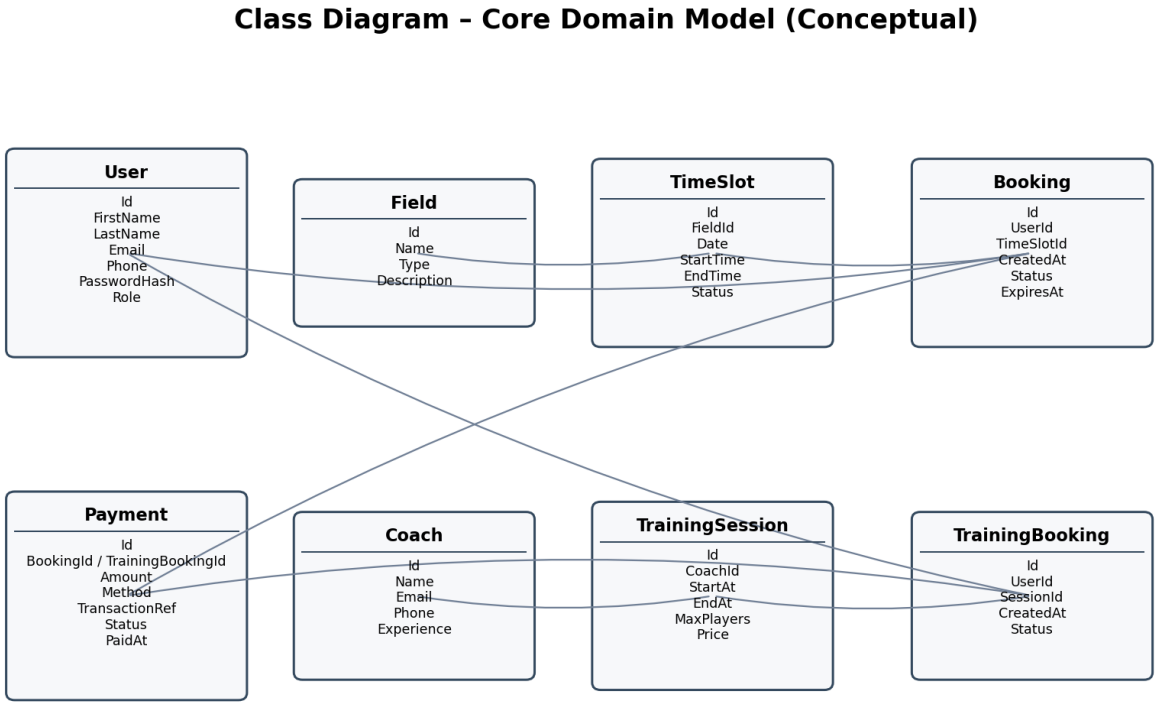


Figure 9. Conceptual Class Diagram – Core Domain Classes

The class diagram is conceptual and focuses on the main entities. Implementation classes such as controllers, DTOs, services, DbContext, and payment webhook handlers can be added in a more detailed technical class diagram.

6. UI/UX Design and Prototyping

6.1 Main Customer Screens

Screen	Main UI Elements	Purpose
Home Page	Club introduction, navigation, field types, training entry point	Helps customers choose a service.
Register / Login	Email, password, registration fields	Authenticates customers.
Field Types	Field type cards, hourly price, details	Displays the club's available field categories.
Field Availability	Date selector, available time slots, consecutive-hour selection	Lets customers choose an available reservation.
Booking Summary	Selected field/time, duration, total price, confirm action	Allows review before creating a pending booking.
Payment Page	Booking reference, amount, payment-provider link/action, status	Directs the customer to complete payment.
Training Sessions	Coach, date/time, duration, price, available capacity	Displays group training options.
Training Booking Summary	Session details, amount, payment action	Confirms the customer's training registration request.
My Bookings	Field and training bookings with statuses	Allows customers to review their records.

6.2 Admin Screens

Screen	Main Functions
Admin Dashboard	Summary cards and charts for bookings and revenue.
Field Management	Create/update fields, field types, descriptions, and prices.

Schedule Management	Create availability dates and time slots; block unavailable periods.
Coach Management	Add, edit, and deactivate coach profiles.
Training Management	Create sessions, assign coaches, set schedules, prices, duration, and capacity.
Booking Management	Review field and training bookings and their statuses.
Payment Management	Review gateway references, amounts, statuses, and verified payments.
Reports	Filter booking and revenue reports by date period.

6.3 UI/UX Guidelines

- Use a consistent navigation structure and clearly distinguish customer pages from the Admin area.
- Show field availability clearly and visually distinguish available, pending, and unavailable slots.
- Display the selected date, duration, and total price before the customer confirms a booking.
- Show a clear payment countdown or expiry time for pending bookings if the team implements a payment deadline display.
- Use clear status labels such as Pending Payment, Confirmed, Cancelled, and Completed.
- Use readable typography, sufficient color contrast, responsive layouts, and clear validation messages.
- Do not rely on color alone to communicate availability or payment status.

6.4 Wireframe Plan

Low-fidelity wireframes should be created for: (1) Customer home page, (2) Field type and availability selection, (3) Booking summary/payment, (4) Training session listing and booking, and (5) Admin dashboard and management forms. The exact visual design and brand colors remain to be selected by the project team.

7. System Deployment and Integration

7.1 Deployment Diagram

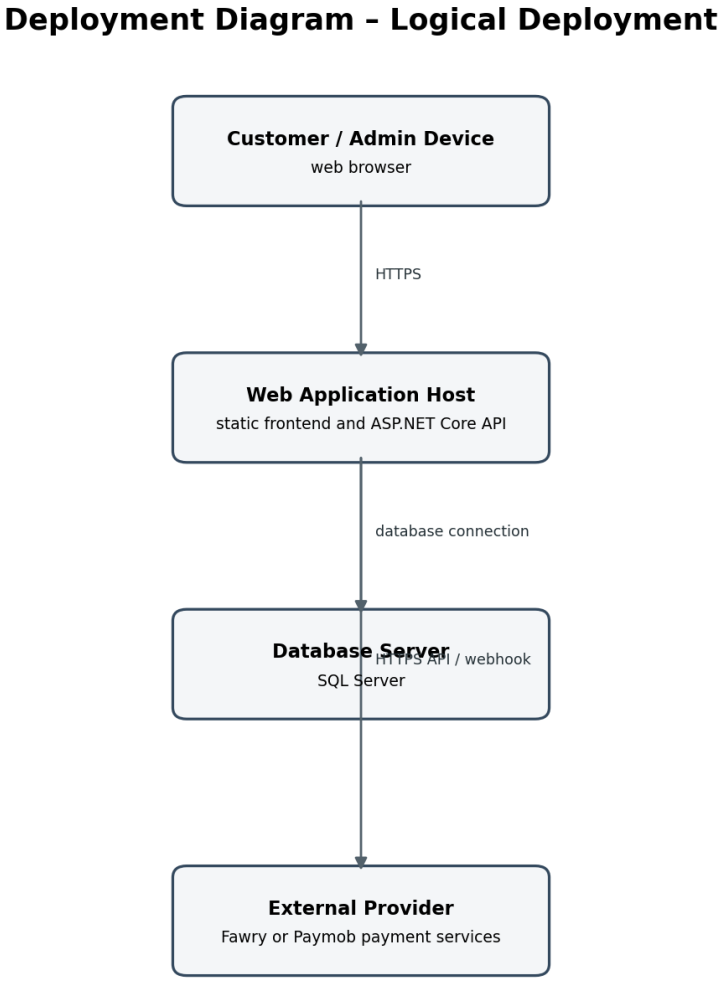


Figure 10. Logical Deployment Diagram

7.2 Component Diagram

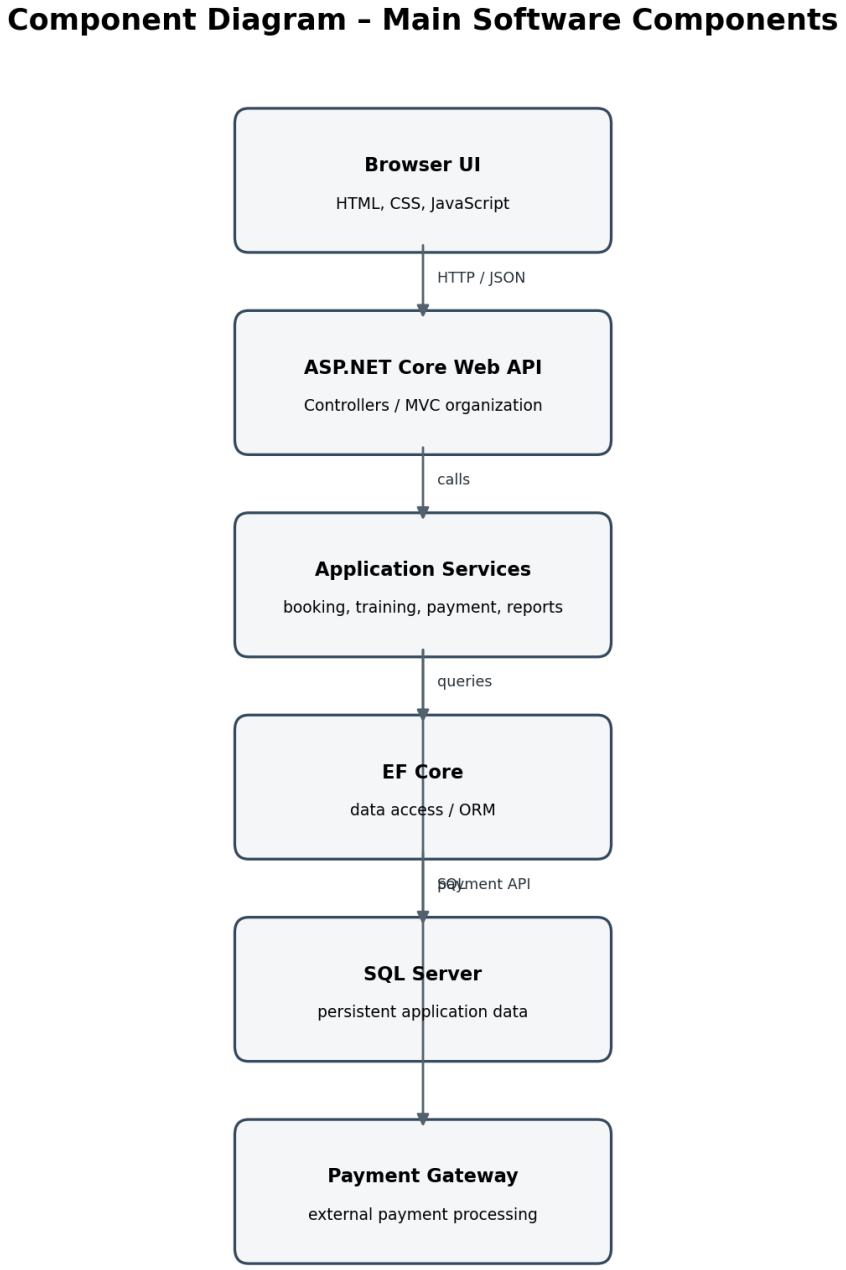


Figure 11. Component Diagram (logical view)

7.3 Integration Description

- The browser communicates with the ASP.NET Core API using HTTPS and JSON.
- The API uses Entity Framework Core to read and update SQL Server records.
- The payment service creates a transaction with the selected provider and stores the provider reference.
- The customer is redirected to the provider's checkout/payment page.
- The provider sends a server-to-server webhook or equivalent payment notification to the API.
- The API validates the notification and, where required, queries the provider to verify transaction status.
- Only a verified successful payment changes the related booking to Confirmed.
- A background scheduled process checks expired Pending Payment bookings and changes them to Cancelled, while releasing their reserved time slots or session capacity.

7.4 Security and Operational Considerations

- Use HTTPS for all public application and webhook endpoints.
- Store secrets and gateway credentials outside source code, using environment variables or a secure secret store.
- Validate webhook signatures or use the provider's recommended verification procedure.
- Use idempotency: processing the same payment notification more than once must not duplicate payment or booking confirmation.
- Use database transactions and concurrency controls when reserving field slots or training capacity.
- Log important payment and booking events without storing sensitive card data.
- Back up the SQL Server database and restrict database access to the application service.

8. Testing Considerations

Test ID	Scenario	Expected Result
TC-01	Customer selects an available field slot.	The system allows the selection.
TC-02	Two customers attempt to book the same field slot.	Only one active reservation succeeds.
TC-03	Customer selects multiple consecutive hours.	The system calculates the combined duration and amount correctly.
TC-04	Customer creates a booking but does not pay before expiry.	The booking becomes Cancelled and availability is released.
TC-05	Gateway confirms successful payment.	The matching booking becomes Confirmed.
TC-06	Gateway sends the same webhook more than once.	The system processes it idempotently without duplicate confirmation.
TC-07	Payment notification contains an incorrect amount/reference.	The system rejects or flags the notification; booking is not confirmed.
TC-08	Customer attempts to book a full training session.	The system prevents the booking.
TC-09	Admin creates a training session.	The session appears in the customer training list when active.
TC-10	Unauthorized customer opens an Admin endpoint.	The API denies access.
TC-11	Admin opens a revenue report.	Only eligible verified payments are included according to the reporting rule.

9. Glossary

Term	Meaning
API	Application Programming Interface; allows the frontend and backend to communicate.
MVC	Model–View–Controller; an architectural organization pattern.
ORM	Object-Relational Mapper; maps application objects to database records.
ERD	Entity-Relationship Diagram; shows database entities and relationships.
DFD	Data Flow Diagram; shows how information moves through a system.
Webhook	A server-to-server HTTP notification sent by an external service when an event occurs.
Confirmed	A booking whose payment has been successfully verified.
3NF	Third Normal Form; a relational database normalization level.